

BOONE, NC, USA

WMO: 722198

Lat: 36.200N

Lon: 81.650W

Elev: 2987

StdP: 13.18

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63819

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.2	(c) 15.1	(d) -4.7	(e) 4.8	(f) 20.9	(g) 0.4	(h) 6.2	(i) 21.8	(j) 35.9	(k) 25.5	(l) 33.5	(m) 27.4	(n) 13.9	(o) 290	(p) 0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.9	(c) 84.1	(d) 67.8	(e) 81.9	(f) 67.3	(g) 80.6	(h) 66.9	(i) 72.3	(j) 77.6	(k) 70.7	(l) 76.6	(m) 69.6	(n) 76.4	(o) 7.3	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.6	130.7	74.2	69.6	121.7	73.6	67.7	113.8	71.7	38.0	78.0	36.8	76.6	35.5	76.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 29.5	(b) 25.0	(c) 20.4	(e) 4.8	(f) 90.6	(g) 5.5	(h) 9.5	(i) 0.8	(j) 97.4	(k) -2.4	(l) 102.9	(m) -5.5	(n) 108.2	(o) -9.5	(p) 115.1		
(a) 29.5	(b) 25.0	(c) 20.4	(e) 3.2	(f) 74.8	(g) 4.9	(h) 6.2	(i) -0.3	(j) 79.3	(k) -3.2	(l) 82.9	(m) -6.0	(n) 86.4	(o) -9.5	(p) 90.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	33.9	36.7	43.1	52.3	60.3	68.1	69.9	68.8	63.4	53.7	43.2	37.9		
	DBStd	14.95	10.83	9.93	9.35	7.82	6.72	4.58	4.14	4.17	5.35	8.16	8.21	8.98		
	HDD50	1887	505	381	253	65	9	0	0	0	1	55	232	386		
	HDD65	4975	963	792	679	382	166	23	11	15	90	359	654	841		
	CDD50	2873	7	9	39	135	330	542	618	582	404	170	28	9		
	CDD65	486	0	0	0	1	22	116	164	131	43	9	0	0		
	CDH74	3051	0	0	0	47	182	742	963	757	284	76	0	0		
	CDH80	474	0	0	0	2	11	113	164	130	23	31	0	0		
(14) Wind	WSAvg	6.3	9.4	9.4	8.2	7.3	5.6	5.0	4.2	3.5	3.4	5.7	6.6	7.9		
(15) Precipitation	PrecAvg	53.2	4.2	3.6	5.4	4.3	4.1	5.0	5.2	5.3	4.6	3.7	3.7	4.1		
	PrecMax	66.1	9.8	9.1	8.8	9.4	9.9	9.7	9.5	11.2	20.3	13.3	8.7	8.1		
	PrecMin	38.5	1.3	1.3	1.7	1.0	1.3	0.9	2.9	1.0	0.3	0.1	0.7	1.1		
	PrecStd	8.2	2.2	1.7	1.8	2.2	2.0	2.0	1.7	2.6	3.9	2.8	2.0	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	65.0	71.7	79.2	81.4	86.1	86.8	86.9	83.7	81.6	69.1	63.4		
		MCWB	53.4	55.0	56.6	57.0	62.4	67.0	67.6	70.1	65.6	69.9	57.0	54.1		
	2%	DB	57.3	60.1	66.2	74.6	78.9	83.0	83.8	83.6	80.6	73.3	64.0	59.7		
		MCWB	51.3	52.1	52.7	56.3	62.2	66.4	68.0	69.1	66.4	61.3	53.4	53.2		
	5%	DB	54.6	56.5	62.8	71.8	75.4	81.1	81.6	81.1	78.3	70.3	61.1	55.7		
		MCWB	49.6	49.8	51.4	56.2	61.3	66.2	68.0	68.3	65.3	59.7	53.4	51.4		
	10%	DB	50.3	52.9	58.5	67.5	73.1	78.9	79.4	78.8	74.6	66.2	57.1	53.5		
		MCWB	46.1	46.5	49.0	54.3	60.8	65.4	68.0	67.7	64.6	59.3	50.8	49.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.7	58.5	58.3	62.5	67.3	71.4	73.9	73.9	74.1	73.0	63.0	59.7		
		MCDB	59.4	62.7	66.4	70.4	75.0	77.3	78.6	80.1	75.0	78.0	63.4	61.1		
	2%	WB	53.5	54.5	56.0	60.1	65.6	70.0	72.1	72.1	69.8	66.3	58.3	55.5		
		MCDB	56.5	58.1	62.4	67.6	72.3	77.9	76.8	78.3	74.6	69.6	61.0	58.0		
	5%	WB	50.6	51.2	54.0	58.5	64.1	68.6	70.5	70.6	68.0	63.6	55.5	52.8		
		MCDB	54.1	55.2	59.2	65.9	70.8	77.1	76.4	76.8	73.8	66.7	58.1	54.7		
	10%	WB	46.4	47.3	51.8	56.7	62.9	67.4	69.4	69.4	66.4	61.1	53.0	49.5		
		MCDB	48.9	51.5	56.8	64.2	69.6	75.8	76.6	75.7	71.7	64.7	56.1	52.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	17.1	18.6	21.4	19.7	18.9	17.9	18.0	18.8	18.9	19.4	16.2		
		MCDBR	20.1	21.9	24.3	28.8	23.7	21.3	20.6	21.3	23.8	24.5	24.0	20.5		
		MCWBR	14.3	15.0	13.2	13.8	10.7	8.9	8.6	9.4	11.2	13.6	14.8	14.8		
	5% WB	MCDBR	17.3	19.4	20.3	21.3	19.9	18.6	17.5	18.0	18.5	18.4	16.8	17.0		
		MCWBR	14.6	15.7	13.6	12.3	10.0	9.0	8.9	10.0	10.7	12.0	13.6	15.0		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.281	0.305	0.354	0.392	0.423	0.457	0.453	0.359	0.317	0.286	0.272		
	taud		2.582	2.561	2.531	2.385	2.282	2.223	2.161	2.147	2.401	2.528	2.594	2.618		
	Ebn at Noon		298	308	307	294	283	273	263	261	284	289	291	292		
	Edh at Noon		23	27	31	37	42	45	48	47	35	28	23	22		
(44) All-Sky Solar Radiation	RadAvg		764	964	1315	1645	1809	1936	1800	1678	1437	1158	873	662		
	RadStd		54	110	86	128	138	134	128	83	130	138	80	61		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+277	+169

Nomenclature: See separate page